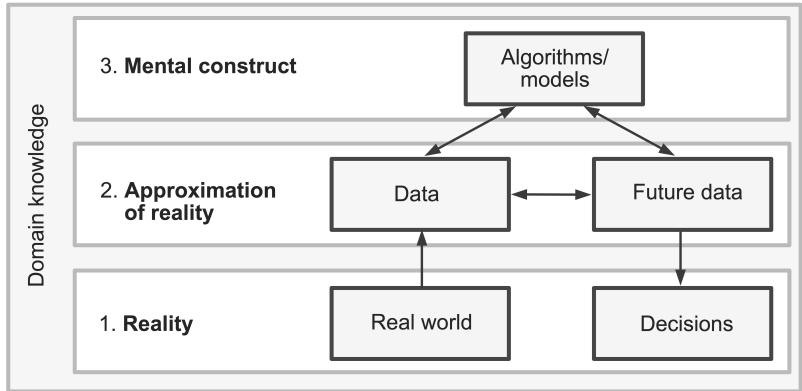


Model diagnostics

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Question: Is our model useful?

Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- ▶ *Sex*: patient's sex (female or male)
- ▶ *Age*: patient's age (in years)
- ▶ *WBC*: white blood cell count
- ▶ *PLT*: platelet count
- ▶ other diagnostic variables. . .
- ▶ *Dengue*: whether the patient has dengue (0 = no, 1 = yes)

Motivating example: Dengue data

Y_i = dengue status (0 = negative, 1 = positive)

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 WBC_i$$

Assumptions and diagnostics

What assumptions do we make in a *linear* regression model?

Logistic regression assumptions

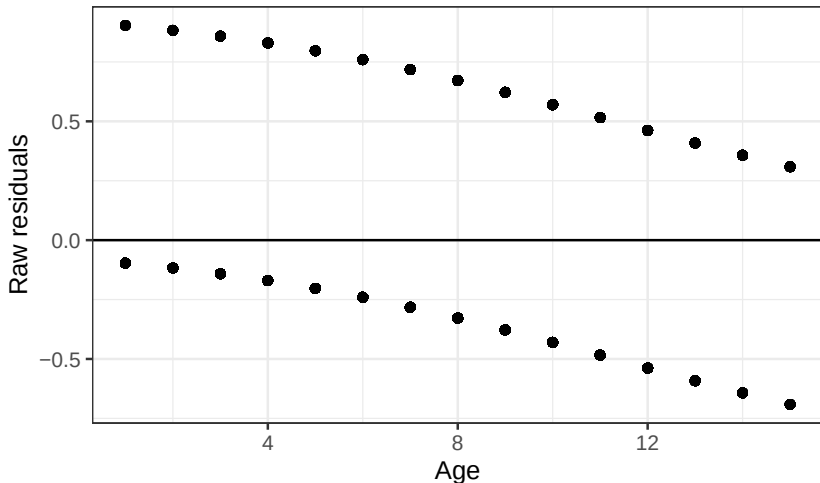
$$Y_i \sim \textit{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \textit{Age}_i$$

Don't use raw residuals for logistic regression

Raw residuals: $Y_i - \hat{\pi}_i$

Example: predicting dengue status with Age



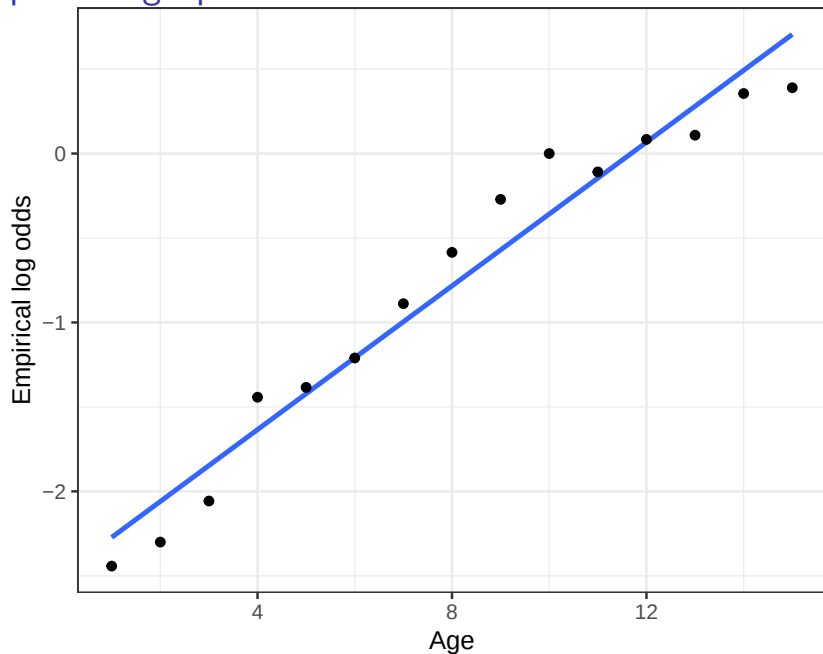
Alternative: empirical logits

For each Age, calculate the *empirical* log odds of dengue (aka *empirical logits*):

```
## # A tibble: 15 x 5
```

##	Age	NoDengue	YesDengue	p	log_odds
##	<int>	<int>	<int>	<dbl>	<dbl>
##	1	1	184	16 0.08	-2.44
##	2	2	419	42 0.0911	-2.30
##	3	3	532	68 0.113	-2.06
##	4	4	457	108 0.191	-1.44
##	5	5	487	122 0.200	-1.38
##	6	6	369	110 0.230	-1.21
##	7	7	360	148 0.291	-0.889
##	8	8	289	161 0.358	-0.585
##	9	9	244	186 0.433	-0.271
##	10	10	185	185 0.5	0
##	11	11	165	148 0.473	-0.109
##	12	12	126	137 0.521	0.0837
##	13	13	96	107 0.527	0.108

Empirical logit plot



Empirical logits with a continuous variable

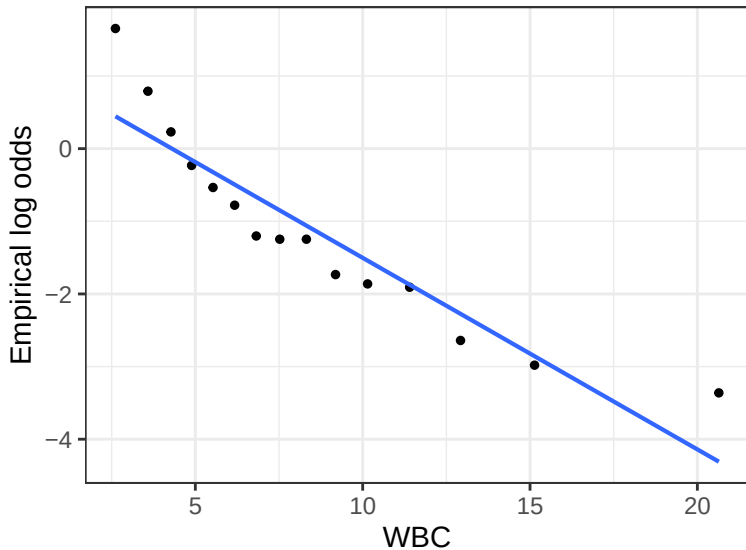
Trying with WBC:

```
## # A tibble: 6 x 5
```

```
##      WBC NoDengue YesDengue      p log_odds
##    <dbl>    <int>    <int> <dbl>    <dbl>
## 1  0.9         0         1      1      Inf
## 2  1.15        0         2      1      Inf
## 3  1.23        0         1      1      Inf
## 4  1.25        0         1      1      Inf
## 5  1.54        0         3      1      Inf
## 6  1.58        0         1      1      Inf
```

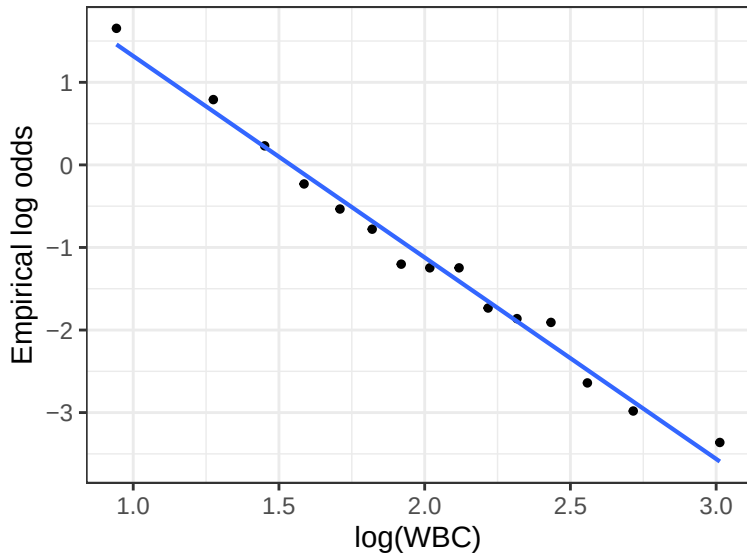
What has gone wrong? What should we do differently?

Binned empirical logit plots

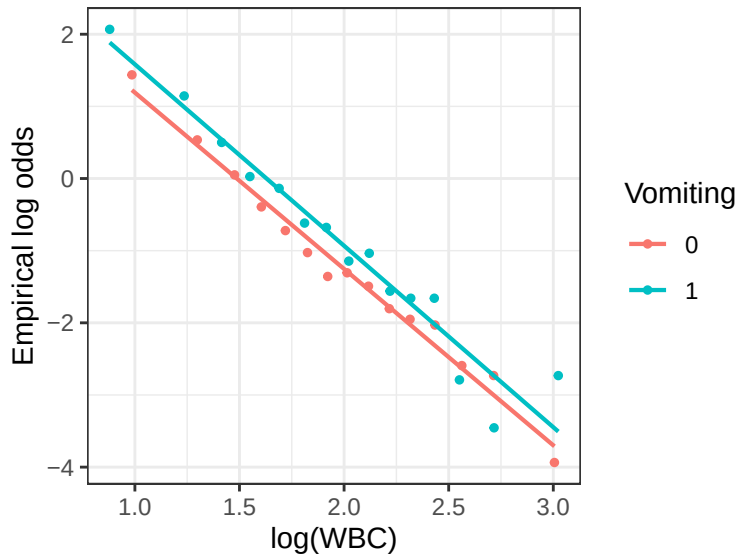


Does the shape assumption seem reasonable?

Binned empirical logit plots



Binned empirical logit plots



Limitations of empirical logit plots

- ▶ Does not assess a fitted model (performed *before* fitting the model)
- ▶ One point per *bin*, not one point per *observation*
- ▶ Can only plot one or two explanatory variables at a time

Class activity

- ▶ Work on the class activity in R (linked on Canvas and the course website)
- ▶ You are welcome to work in groups
- ▶ At the end of class, submit HTML file on Canvas (one per group, listing all group member names)